Spatially Enhanced Refined Classifier for Cross-Scene Hyperspectral Image Classification

Zhengshan Wang, Long Chen, *Senior Member, IEEE*, Yifei Tian, Juan He, and C. L. Philip Chen, *Life Fellow, IEEE*

***Abstract*— Conventional models for hyperspectral image (HSI) classification usually demand a substantial quantity of labeled training data. However, when labeled training HSIs are unavailable or have different distributions from test HSIs, many classification models tend to exhibit significant performance declines. For the cross-scene HSI classification task, unsupervised domain adaptation (UDA) technique has been widely developed. Existing discrepancy-based or adversarial-based UDA methods may fail to learn discriminative class boundaries when large class distribution shift (CDS) exists. To alleviate this limitation, we propose the model named spatially enhanced refined classifier (SERC), which includes a coarse classifier (CC) and a refined classifier (RC). The RC constructs a memory module to fuse global-spatial and spectral information simultaneously and uses the neighborhood aggregation technique to generate the refined predictions. The refined predictions are then transferred to pseudolabels to train the CC. Thereby, a mutually reinforcing relationship between the two classifiers is established. Furthermore, we propose the class distribution match (CDM) strategy to further alleviate the serious CDS problem. Notably, SERC does not require additional training parameters, which are commonly used in existing UDA methods. Despite its simplicity, SERC achieves outstanding results. Our method has been extensively evaluated on three public HSI datasets and has shown superior performance compared with state-of-the-art (SOTA) approaches. The source code can be found at https://github.com/Wangzs0228/SERC.**

***Index Terms*— Class distribution shift (CDS), domain adaptation (DA), hyperspectral image (HSI) classification, pseudolabels.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are captured across multiple spectral bands, allowing for a detailed analysis of the reflectance properties of various objects or materials [1], [2]. Unlike the natural visible light image, the HSI offers a richer spectral information within each pixel and has been extensively applied across various domains recently, including urban development, environmental detection, agronomy, military applications, and so on [3]. Unlike in the natural visible light image, the classification task in the HSI will assign category labels to all pixels, which is more similar to the traditional segmentation task [4], [5], [6], [7], [8], [9], [10], [11].

Although deep learning models have demonstrated significant success [12], they often encounter challenges when tested in domains different from the training set [13], [14]. This domain shift is often caused by variations in capture conditions, including atmospheric conditions, soil moisture, lighting conditions, and other factors [15]. Moreover, the process of annotating HSI data will consume much time and requires significant labor, leading to a scarcity of labeled train data [16]. Given this background, the utilization of domain adaptation (DA) [13], [17] has become prevalent. Specifically, unsupervised DA (UDA) is a subfield of DA that specifically addresses scenarios where the target domain (TD) lacks any labeled data and has been extensively used in the task of cross-scene HSI classification [4], [5], [6], [7].

The existing UDA methods can be typically classified into three types [13]: 1) the instance-based methods reweight the samples from the source domain (SD) to match TD distributions [14]; 2) the classifier-based methods uses unlabeled samples of TD to optimize classifiers trained on SD [18]; and 3) the feature-based methods adjust the feature space of the SD and TD so that the feature distribution of the different domains is similar, which can be further divided into discrepancy-based and adversarial-based methods. For example, Zhang et al. [17] proposed topological structure and semantic information transfer network (TSTnet), which combines a graph convolutional network (GCN) with a convolutional neural network (CNN) and uses maximum mean discrepancy (MMD) to align distributions. Zhang et al. [19] proposed a locality robust DA (LRDA) method to reduce the domain-shift between the SD and TD. Zhu et al. [5] introduced the deep subdomain adaptation network (DSAN), which uses a local

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MMD (LMMD) to align the distributions of domain-specific layer activations across different domains. Huang et al. [20] introduced the two-branch attention adversarial DA (TAADA) network, which mitigates distribution discrepancies through a two-classifier discriminator design. Huang et al. [21] proposed spatial–spectral weighted adversarial DA (SSWADA) for the cross-scene wetland problem. The key idea is using weighted adversarial discrimination to align the feature distribution of source and target scenes. Zhao et al. [22] proposed UDA model named toward multilevel features and decision boundaries (ToMF-B), which can align task-related features and learn task-specific decision boundaries in parallel based on the maximum classifier discrepancy (MCD), Feng et al. [23] proposed a class-aligned and class-balancing generative DA (CCGDA) method, which designs a sampler to resample the TD data in an adaptive proportion according to the statistical calculation through confidence and distribution of pseudolabels.

While significant progress has been made, limitations still exist in these works. First, feature alignment through adversarial learning can unintentionally diminish the discriminability of features [24]. Second, the class distribution shift (CDS) problem is more pronounced in cross-scene HSI classification than in natural image tasks, making it more challenging for models to learn discriminative class boundaries [25], [26], [27]. In this case, overly enforcing alignment between two distinct domains may cause the model to learn distorted domain-invariant features. Finally, due to the absence of pretrained feature extractors [28], [29], [30], although discrepancy-based or adversarial-based UDA methods can project the features of two domains into the same space, these features may not necessarily transferable due to the lack of prior knowledge support. At worst, deep models after UDA may even perform worse than original models [3]. Therefore, the self-training or pseudolabeling technique [31], [32] that can leverage the abundant unlabeled data to improve performance has become a promising approach [25], [26].

However, the serious domain shift in cross-scene HSI can lead to unreliable predictions when applying a classification model trained on the SD to the TD [31]. In this article, these unreliable predictions are referred to as coarse predictions. Fortunately, unlike natural visible light images, HSIs contain additional spatial coordinates in addition to spectral information. These coordinates can provide valuable global-spatial information. According to Tobler's first law of geography [33], all attribute values on a geographic surface are interrelated, with closer values being more strongly related than those that are farther apart. Therefore, we propose spatially enhanced refined classifier (SERC), which can incorporate the coordinate spatial information to refine the coarse predictions. To be specific, SERC consists of two classifiers: a coarse classifier (CC) and a refined classifier (RC), which both share the same feature extractor. The CC is trained using both the data from SD with ground-truth labels and the data from TD with pseudolabels. Then, it will output the coarse predictions. The RC is a parameter-free classifier based on the nearest neighbor strategy that is primarily designed for the TD. It uses prior knowledge from the CC to construct a memory module. This module effectively stores information regarding spectral features, global-spatial coordinates, and coarse predictions. First, the RC uses the spectral and global-spatial feature to identify neighbors for each data in TD. Then, by aggregating coarse predictions from these neighbors, it generates the refined predictions. Finally, the refined predictions are transferred to pseudolabels, which will aid the training of the CC. In this way, the CC and RC establish a mutually reinforcing relationship, and ultimately, the RC is responsible for generating predictions during the inference stage. The existing self-training approaches, such as [18] and [34], typically construct class prototypes and compute the similarity between TD samples and class prototypes to generate pseudolabels for them. This strategy overlooks the spatial characteristics of the samples. Therefore, our proposed method can obtain better results by aggregating the information of neighborhoods in both the spatial and spectral space.

Current UDA methods primarily concentrate on aligning features of different domains. Unfortunately, such inadequate consideration is not enough to guarantee effective and beneficial adaptation. As a nonnegligible factor in domain shift, CDS refers to the disparity in class distributions between the SD and TD [35]. A significant disparity in CDS can greatly impact the transfer effectiveness of the model. In existing works, to prevent the generators from mode collapse caused by category imbalance, Feng et al. [23] set a data sampler to balance the ratio of different categories in data, so that the model has roughly the same predictive bias for each category. However, the class balancing approach in CCGDA is relatively simplistic, as it merely ensures that each category has an equal proportion in the dataset during model training. Since the proportions of different classes in the TD may be imbalanced, ensuring equal ratios for each class during training still cannot guarantee the model can adapt effectively to the TD. Our work is a further extension of them. Rather than simply balancing the ratio for different classes, we aim to make the class distribution of the SD data closely mimic that of the TD. To be specific, we propose class distribution match (CDM) based on the historical category predictions of the two domains. CDM can fully leverage the intraclass coupling relationship [36] and help the CC learn a class distribution more similar to the TD, which can further improve the performance of SERC. Compared with the existing spatial-information-based methods [17], [34], [37], which calculate distances between pixels and input the graph structured data into the GCN for more stable feature extraction, our approach avoids increasing model complexity and dependency on the adjacency matrix quality. This enhances generalizability and feasibility while keeping the neural network parameters streamlined and effective.

In summary, this article presents a novel self-training UDA method for cross-scene HSI task, using neighborhood information to enhance the generation of pseudolabels. Different from the existing adversarial feature alignment methods or complex UDA approaches that use dual classifiers [18], [38], [39] or GCN networks [17], [34], our model architecture is highly streamlined. In addition, we address the common issue of CDS in the cross-scene HSI task, enabling the model to adapt more effectively to the TD and further improving the performance of our proposed method. Comprehensive experiments on three

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publicly available benchmarks of HSIs have been conducted to demonstrate the superiority of our proposed SERC. The main contributions include the following.

1) We propose a novel self-training UDA model for cross-scene HSI problem called SERC. Extensive evaluations demonstrate that SERC can get excellent performance on three public datasets.
2) SERC contains two classifiers named CC and RC. It can generate more accurate pseudolabels by aggregating coarse predictions from neighboring samples, considering their spectral and global-spatial characteristics.
3) Aiming at the serious CDS in the cross-scene HSI classification task, we propose the CDM to help the CC learn a class distribution more similar to the TD.

This article is structured as follows: Section II provides an overview of related works. Section III presents the proposed methodology of the SERC model. Section IV describes the experimental datasets, setup, analysis, and discussions. Finally, Section V summarizes the conclusion and further works.

II. Related Work

A. Unsupervised Domain Adaptation in Cross-Scene Hyperspectral Image Classification

In the cross-scene HSI classification task, the SD and TD have distinct yet similar distributions [3]. This distribution discrepancy is typically caused by variations in sensors, scenes, acquisition conditions (atmospheric, illumination, or acquisition angle), and acquisition time. Consequently, when the model trained in one scenario is tested in another unseen scenario, there will be a significant drop in performance. UDA aims to address this problem without requiring extensive manual effort [13], [15]. The existing UDA works can be broadly categorized into discrepancy-based and adversarial-based methods [13]. There are also some works that used the self-training or pseudolabeling technique. For instance, minimum class confusion (MCC) [40] focuses on minimizing pairwise class confusion in the TD. To improve the quality of pseudolabels and address classifier bias, auxiliary target domain-oriented classifier (ATDOC) [41] introduces an auxiliary classifier specifically for the TD.

Compared with UDA in natural visible light images, UDA in the HSI is more challenging due to its pixel-level input and output, as well as its complex spatial information. For example, Zhang et al. [17] proposed TSTnet, which uses a graph structure to capture topological relationships. Cheng et al. [37] introduced prototype adversarial and dual adversarial strategies to enhance feature separability by increasing the distance between class prototypes. Their common characteristic is that they all rely on feature alignment. However, feature alignment in the HSI can be unstable when large CDS exists or without a pretrained model [25], [27]. To tackle this issue, we use the self-training or pseudolabeling technique in this article [31], [32] to tackle the cross-scene classification task in HSI.

B. Pseudolabeling in Unsupervised Domain Adaptation

As a heuristic approach in machine learning, pseudolabeling technique is widely used in semi-supervised learning [13]. As a typical example, high-confidence predictions on unlabeled data are gradually treated as pseudolabels [42]. To improve the quality of pseudolabels, Iscen et al. [43] used a transductive label propagation method based on the manifold assumption to generate pseudolabels. Zou et al. [44] introduced a class-balanced self-training framework to prevent the gradual dominance of large classes during pseudolabel generation.

Most existing UDA works that use the pseudolabeling technique primarily concentrate on natural visible light images, with limited research dedicated to HSIs. In this context, Cheng et al. [34] proposed using class centroids to store the spectral information of class prototypes and construct a virtual classifier for HSIs. However, these class centroids can only provide a coarse representation of the spectrum structure but fail to capture the global-spatial structure effectively. To address this limitation, inspired by the concept of passing messages through neighbors [41], we design the RC based on the nearest neighbor strategy, leveraging the benefits of considering both the spectral and global-spatial characteristics. Furthermore, we also adjust the training weights of different categories according to the class distribution of the SD and TD. The method in this study is very simple and effective, and the test results on multiple datasets prove its excellent performance.

C. Feature Extractor for Hyperspectral Image Classification

For the cross-scene HSI classification task, the input for feature extractor is the multiband spectral information, the pretrained model like ResNet [28] or transformer [29], [30] commonly used in natural visible light images cannot be directly used here. When developing UDA methods for HSI, the preprocessing of data and the design of a suitable feature extractor are crucial. These steps have a significant impact on the effectiveness of the UDA approach. Among the existing works, HSI training data are mainly input in two forms: single pixel and pixel block. The former takes each pixel as input, while the latter takes pixel patches composed of the central pixel and its surrounding pixels as input. For example, the ith pixel block is centered at x_i and contains neighboring pixels of size $n \times n$ around it. In general, the input in the form of pixel blocks can often achieve better results due to its inclusion of both spectral information and the connectivity between neighboring pixels. Moreover, the current feature extractors commonly used include 1-D-CNN, 2-D-CNN, 3-D-CNN, GCN, and fully convolutional networks (FCN) [45]. Among them, 3-D-CNN is particularly effective in leveraging both the spectral and local-spatial information. In this study, we use the following strategy: the original HSI data will be preprocessed into pixel blocks, and a 3-D-CNN based architecture will be used as the feature extractor. Subsequently, the feature extractor will undergo training process for the cross-scene HSI classification task. The detail structure of the feature extractor will be given in the experimental section.

D. Definition of Class Distribution

To avoid reading obstacles, we provide the definitions of class distribution and CDS mentioned in this article.

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1) Class Distribution: Class distribution refers to the statistical distribution of class labels in a dataset. It characterizes how frequently each class label appears within the dataset, typically represented as a probability distribution across the distinct classes. For a dataset with K classes, the class distribution can be denoted as

$$P(\mathcal{D}) = \{r_1, r_2, \ldots, r_K\} \tag{1}$$

where r_k represents the ratio of instances in the dataset $\mathcal{D}$ that belong to class k and satisfies $\sum_{k=1}^{K} r_k = 1$.

2) Class Distribution Shift: CDS occurs when the class distribution of an SD differs from that of a TD. This shift can impact the performance of machine learning models trained on the SD when applied to the TD. For example, CDS can be quantitatively described as the divergence between the two distributions

$$\text{CDS} = D(P(\mathcal{D}_s)||P(\mathcal{D}_t)) \tag{2}$$

where $P(\mathcal{D}_s)$ is the class distribution of the SD and $P(\mathcal{D}_t)$ is the class distribution of the TD. The divergence D can be measured using various metrics, such as Kullback–Leibler divergence or Jensen–Shannon divergence. A significant CDS often leads to challenges in classification tasks, as models may not generalize well across domains due to the differing availability or prevalence of ground-truth labels.

III. Proposed Spatially Enhanced Refined Classifier Approach

This section provides a detailed description of the structure of the proposed SERC model. In addition to the standard task classifier (also called CC in this article), SERC includes an extra RC aiming to use the global-spatial structure of the TD and output more accurate predictions.

A. Problem Preliminary

Define the SD $\mathcal{D}_S$ as $\{(x_i^s, \theta_i^s, y_i^s)\}_{i=1}^{n_s}$ with n_s samples, and the TD $\mathcal{D}_T$ as $\{(x_i^t, \theta_i^t, y_i^t)\}_{i=1}^{n_t}$ with n_t samples. Here, $x_i^s, x_i^t \in \mathbb{R}^b$ represent the ith labeled data sampled from $\mathcal{D}_S$ or $\mathcal{D}_T$, respectively. Both of them have the dimensionality of b bands. y_i^s denotes the ground-truth label, which covers K classes totally. $\theta_i^s \to \{\theta_{h,i}^s, \theta_{v,i}^s\}$ and $\theta_i^t \to \{\theta_{h,i}^t, \theta_{v,i}^t\}$ are the spatial coordinates, where $\theta_{h,i}^s$ and $\theta_{h,i}^t$ represent the horizontal axis positions of data in the SD and TD, respectively, and $\theta_{v,i}^s$ and $\theta_{v,i}^t$ represent the vertical axis positions of data in the SD and TD, respectively. The deep classification model can be decomposed into a feature extractor $G(\cdot)$ that outputs feature $f_i \in \mathbb{R}^d$, i.e., $f_i = G(x_i)$, and a task classifier $C(\cdot)$ that generates the corresponding $\hat{y}_i \in \mathbb{R}^K$, i.e., $\hat{y}_i = C(f_i)$. The corresponding loss function is shown as

$$\mathcal{L}_{\text{cls}}(x^s, y^s) = \mathbb{E}_{(x_i^s, y_i^s)\sim\mathcal{D}_S} \mathcal{L}_{\text{ce}}\left(C\left(G\left(x_i^s\right)\right), y_i^s\right) \tag{3}$$

where $\mathcal{L}_{\text{ce}}(\cdot,\cdot)$ is the cross-entropy loss function. The goal of UDA is to train the feature extractor and task classifier using data from the SD, and the trained model can perform well in the TD.

B. Spatially Enhanced Target Classifier

The structure of the proposed SERC model, as depicted in Fig. 1, consists of two parts, i.e., the CC and the RC. The CC will output features and coarse predictions for both the domain data, and the features and coarse predictions related to the TD will be stored in the memory module. The RC generates pseudolabels $\widehat{p}_i$, which are then used to train the CC. The feature extractors of the two classifiers are parameter-shared. In this section, we will introduce them in detail.

To prevent trivial sample selection and alternating training, SERC uses a memory module denoted as $x^m \to \{f^m, \theta^m, \hat{y}^m\}$, which is used to store information of all target samples. To be specific, the memory module is composed of three memory banks (MBs) named the feature MB f^m, the coordinate MB θ^m, and the prediction MB $\hat{y}^m$, which are build to store the features, coordinates, and coarse predictions, respectively. Here, we use x_j^m to represent the jth sample in the memory module. To avoid ambiguity, the sharpen strategy will be used [46]. The sharpened prediction of $\hat{y}_j$ can be calculated as $\hat{y}_{j,k}^m = \hat{y}_{j,k}^{(1/T)} / \sum_k \hat{y}_{j,k}^{(1/T)}$. Here, $\hat{y}_{j,k}$ is the value of the kth class in $\hat{y}_j$. As the temperature T approaches 0, the probability will converge to a point mass [42]. In this article, we set T to 0.5. Then, the L2-normalized feature vector $f_j^m = G(x_j^t)/\|G(x_j^t)\|_2$, the coordinate $\theta_j^m = \{\theta_{h,j}^t, \theta_{v,j}^t\}$, and the sharpened prediction $\hat{y}_j^m$ are written in the memory module based on their indexes. The memory module will keep updating every mini-batch during training. Based on the memory module, an RC based on the nonparametric neighborhood aggregation strategy can be constructed.

First, the distance $d(x_i^t, x_j^m)$ between x_i^t and x_j^m is calculated based on the fusion of $d^f(x_i^t, x_j^m)$ and $d^\theta(x_i^t, x_j^m)$. The former is the cosine similarity between $G(x_i^t)$ and f_j^m, and the latter is the Euclidean distance between $(\theta_{h,i}^t, \theta_{v,i}^t)$ and $(\theta_{h,j}^m, \theta_{v,j}^m)$

$$d^f\left(x_i^t, x_j^m\right) = 1 - \frac{G\left(x_i^t\right)^\top f_j^m}{\left\|G\left(x_i^t\right)\right\|_2 \left\|f_j^m\right\|_2} \tag{4a}$$

$$d^\theta\left(x_i^t, x_j^m\right) = \left(\theta_{h,i}^t - \theta_{v,j}^m\right)^2 + \left(\theta_{h,i}^t - \theta_{v,j}^m\right)^2 \tag{4b}$$

$$d\left(x_i^t, x_j^m\right) = d^f\left(x_i^t, x_j^m\right) + \lambda_1 d^\theta\left(x_i^t, x_j^m\right) \tag{4c}$$

where λ_1 is the hyperparameter used to balance the influence of spectral and global-spatial information. The samples x_j^m in the memory module are sorted in descending order according to their distances $d(x_i^t, x_j^m)$. Subsequently, several samples in the memory module with the minimum distances are selected as neighbors of x_i^t. The number of neighbors is denoted as λ_0. It should be noted that x_i^m cannot be selected as a neighbor of x_i^t and needs to be eliminated. Here, $\mathcal{N}_i^m$ denotes the index set of these neighbors for the data x_i^t. Then, the coarse predictions of these neighbors in the prediction MB can be aggregated to represent the refined prediction of x_i^t.

Using average aggregation to generate refined predictions is an intuitive approach. However, the distance between x_i^t and different neighbors can be different, indicating differences in their contributions. Moreover, these neighbors may have varying levels of confidence based on their information richness. Therefore, treating them equally is not reasonable. To generate more accurate refined predictions, the aggregation process in

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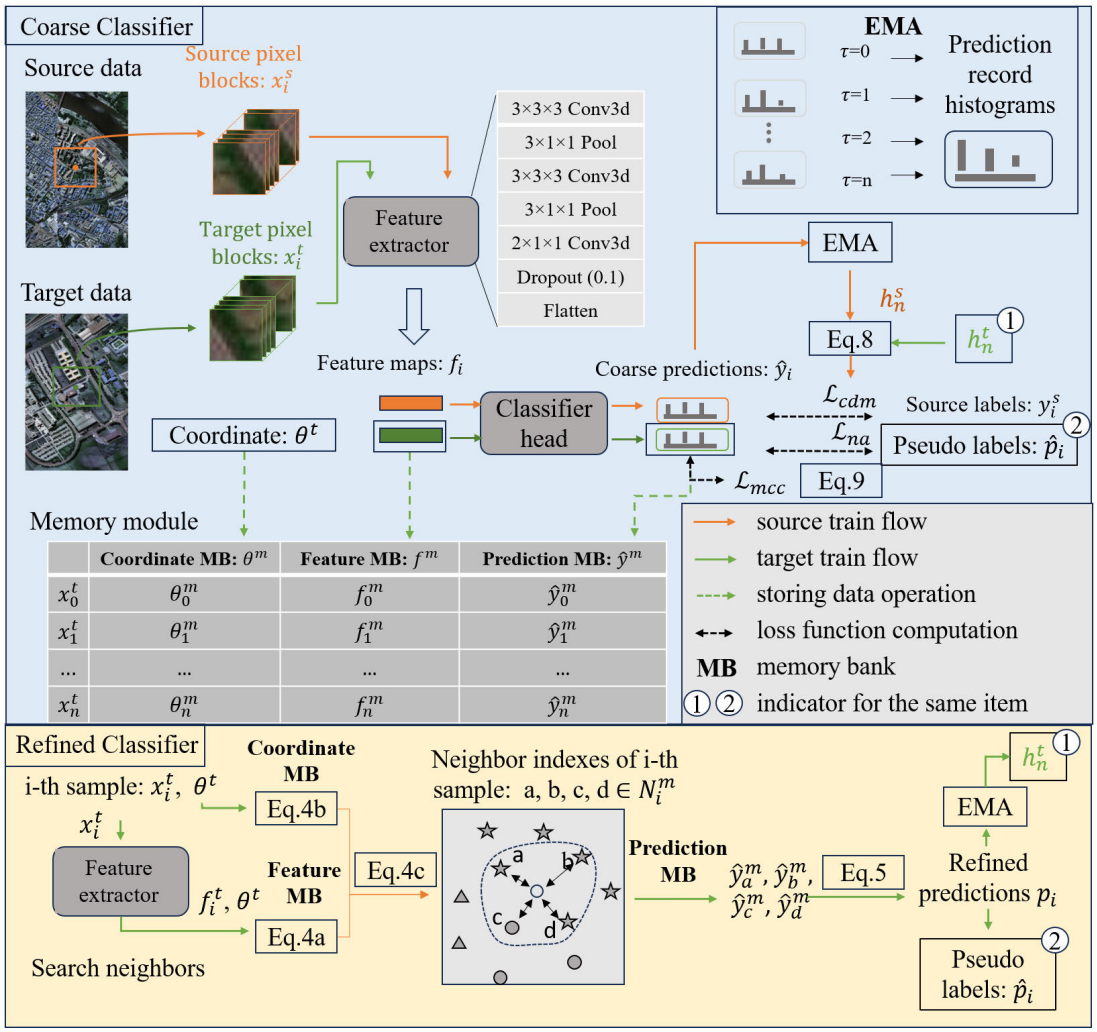


Fig. 1. Total structure of the proposed SERC. MB is the abbreviation for memory bank. Solid lines indicate the direction of data flow. The orange line represents operations related to the SD, while the green line corresponds to the TD. The black dashed line illustrates the computation of the loss function, and the green dashed line signifies the operation of storing data in the MB. A more detailed description of this figure can be found in the beginning of Section III-B.

this article takes into account both the proximity of each neighbor measured by distances and their informative level measured by entropy measurement

$$\text{Entropy}\left(\hat{y}_j^m\right) = -\sum_{k=1}^{K} \hat{y}_{j,k}^m \log \hat{y}_{j,k}^m \tag{5a}$$

$$p_i = \frac{1}{\lambda_0} \sum_{j \in \mathcal{N}_i^m} \frac{\left(1 + e^{-\text{Entropy}\left(\hat{y}_j^m\right)}\right)}{d\left(x_i^t, x_j^m\right)} \hat{y}_j^m. \tag{5b}$$

After getting the refined prediction p_i, the final hard pseudolabel $\hat{p}_i$ can be obtained by selecting the category index with the highest probability prediction, represented as $\hat{p}_i = \arg\max_k p_{i,k}$. Here, $p_{i,k}$ denotes the value in p_i corresponding to the kth class. Finally, a cross-entropy loss is imposed on the TD as

$$\mathcal{L}_{\text{na}} = \mathbb{E}_{(x_i^t, y_i^t) \sim \mathcal{D}_T} \mathcal{L}_{\text{ce}}\left(C\left(G\left(x_i^t\right)\right), \hat{p}_i\right). \tag{6}$$

It is important to note that our strategy takes into account the global structure beyond regularization within a mini-batch, similar to the approach in [40] and [47]. The training and testing phases of the proposed SERC can be summarized as follows: During the training phase, pixel blocks from the SD and TD are input into the CC. Simultaneously, the data in the memory module are continuously updated according to the output of CC. In the RC, the neighbors of each TD data are identified in the spatial–spectral space. Then the refined predictions can be obtained through aggregating the coarse predictions of these neighbors in the prediction MB. During the testing phase, the saved information in the memory module will be frozen, and the RC is used to evaluate the model's performance in the TD.

C. Class Distribution Match

CDS is a significant issue observed in the cross-scene HSI classification task. For instance, in the Pavia dataset, the University of Pavia (UP) is commonly used as the SD, while the Pavia Center (PC) serves as the TD. In the UP scene, there are 947 instances of the "shadow" category and 18 649 instances of the "grass" category. In contrast, the PC scene contains approximately equal numbers of these two classes, with 2863 instances of "shadow" and 3090 instances of "grass." The significant presence of CDS underscores the

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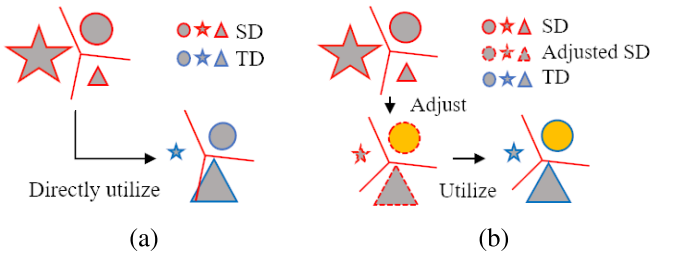


Fig. 2. Schematic illustration on DA without and with CDM. The SD and TD are distinguished by the color of the circle borders. Each of the shapes represents a different category, with the size indicating the number of samples in that category. The decision boundary learned by the classifier is represented by straight lines. (a) Decision boundary trained in the SD will perform poorly in the TD with large CDS. (b) Adjusted SD will have a similar class distribution as the TD, and the decision boundary trained on the adjusted SD will show better results in the TD.

importance of addressing this issue in the cross-scene HSI classification task.

The CDS exhibits pernicious effects in the cross-scene HSI classification task. To be specific, when there are significant variations in class distributions, the decision boundary of the classifier trained on the labeled SD may not be appropriate for the TD. Therefore, our objective is to directly adjust the weights of different categories to enhance similarity of class distribution between SD and TD, as depicted in Fig. 2. Since different classes have different levels of intraclass diversity, some classes are harder to classify than others [36] or different classes may also share similar characteristics. We will not measure the class distribution by calculating the number of samples for each class. On the contrary, we believe that neural networks possess the capability to dynamically analyze this coupling, which will be manifested in their output. Therefore, we propose CDM based on the historical category predictions of the two domains.

Specifically, we compute the expectation of the model's predictions on each class to estimate the class-specific learning status via the exponential moving average (EMA). The prediction record histograms of the SD and TD are represented by h_τ^s and h_τ^t, respectively. They are two lists containing $[h_{\tau,1}^s, h_{\tau,2}^s, \ldots, h_{\tau,K}^s]$ and $[h_{\tau,1}^t, h_{\tau,2}^t, \ldots, h_{\tau,K}^t]$, respectively. The h_τ^s and h_τ^t will be used to approximate the class distribution of the SD and TD. τ denotes the τth training iteration. Then, a fine-grained class-specific rescale is calculated here

$$h_{\tau,c}^s = \begin{cases} \dfrac{1}{K}, & \text{if } \tau = 0 \\ \lambda h_{\tau-1,c}^s + (1-\lambda)\dfrac{1}{B}\displaystyle\sum_{i=1}^{B} \hat{y}_{i,c}^s, & \text{otherwise} \end{cases} \tag{7a}$$

$$h_{\tau,c}^t = \begin{cases} \dfrac{1}{K}, & \text{if } \tau = 0 \\ \lambda h_{\tau-1,c}^t + (1-\lambda)\dfrac{1}{B}\displaystyle\sum_{i=1}^{B} p_{i,c}^t, & \text{otherwise} \end{cases} \tag{7b}$$

where the parameter λ determines the update speed and is set to 0.999 in this study. B is the batch size. The $h_{\tau,c}^s$ and $h_{\tau,c}^t$ represent the recorded values of category c in the τth iteration for the SD and TD, respectively. Then, we can calculate the specific category weights, as shown as

$$w_{\tau,c} = \frac{h_{\tau,c}^t}{h_{\tau,c}^s} \tag{8a}$$

$$\mathcal{L}_{\text{cdm}}(x^s, y^s) = \mathbb{E}_{(x_i^s, y_i^s)\sim\mathcal{D}_\mathcal{S}} w_i^s \mathcal{L}_{\text{ce}}\big(C\big(G\big(x_i^s\big)\big), y_i^s\big) \tag{8b}$$

where $w_{\tau,c}$ is the weight of class c in the τth iteration, w_i^s is the corresponding weight of the class where x_i^s belongs to. It should be noted that for the sake of simplicity, the expression of w_i^s has omitted τ, but its value will continuously change throughout the training process. The CDM enhanced cross-entropy loss is denoted as $\mathcal{L}_{\text{cdm}}$. To further regularize the training process, we also introduce the MCC term proposed by Jin et al. [40]. This regularization term, denoted as $\mathcal{L}_{\text{mcc}}$, is incorporated into the overall training objective. In summary, the total loss function used for optimizing the classification model can be expressed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cdm}} + \lambda_2 \sigma \mathcal{L}_{\text{na}} + \mathcal{L}_{\text{mcc}} \tag{9}$$

where σ is the training progress linearly changing from 0 to 1, and λ_2 is a hyperparameter controlling the effect of $\mathcal{L}_{\text{na}}$. The coefficient $\lambda_2\sigma$ is a positive parameter that reflects the learning status of the model. It is expected to increase gradually during training as the model gains more confidence. For all the cross-entropy losses in this article, we will use the label-smoothing regularization [48].

IV. Results and Discussion

We will conduct experiments to validate the performance of the proposed SERC using three cross-scene HSI datasets: Houston, Pavia, and Shanghai–Hangzhou (S–H). The selected models used for comparison in this article are: deep adaptation network (DAN) [50], domain-adversarial training of neural networks (DANN) [51], DSAN [5], MCD [38], confident learning-based domain adaptation (CLDA) [39], TSTnet [17], SSWADA [21], and CCGDA [23]. Among these, DAN, DANN, DSAN, and MCD are classic UDA algorithms, while CLDA, TSTnet, SSWADA, and CCGDA represent some of the latest UDA works for the cross-scene HSI classification task. Furthermore, class-specific accuracy (CA), overall accuracy (OA), and the kappa coefficient (KC) are used to assess the proposed model.

A. Experimental Data

1) Houston Dataset: The Houston dataset used in this study consists of two scenes, Houston 2013 [52] and Houston 2018 [53], which were captured by different sensors in different years. These scenes cover the University of Houston campus and its surrounding areas. The Houston 2013 dataset comprises an image with dimensions of 349 × 1905 pixels. It contains 144 spectral bands ranging from 380 to 1050 nm, with a spatial resolution of 2.5 m. The Houston 2018 dataset has the same wavelength range but consists of 48 spectral bands. Both the Houston 2013 and Houston 2018 scenes share seven consistent classes. For our analysis, we extract 48 spectral bands (wavelength range of 0.38–1.05 μm) from the Houston 2013 scene that corresponds to the Houston 2018 scene. The list of classes and the number of samples are presented in Table I. Furthermore, the pseudocolor images and ground-truth maps can be observed in Fig. 3.

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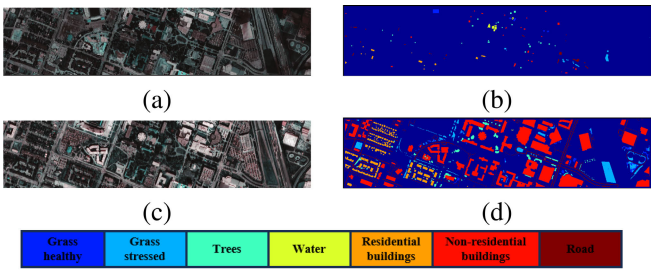


Fig. 3. Pseudocolor image and ground-truth map of the Houston dataset. (a) Pseudocolor image of Houston 2013. (b) Ground-truth map of Houston 2013. (c) Pseudocolor image of Houston 2018. (d) Ground-truth map of Houston 2018.

TABLE I
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HOUSTON DATASET

Class		Number of Samples	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE PAVIA DATASET

Class		Number of Samples	
No.	Name	UP (Source)	PC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

2) Pavia Dataset: The Pavia dataset consists of two scenes: UP and PC. The PC scene has dimensions of 1096 × 715 pixels and contains 102 spectral bands. The UP scene has 103 spectral bands, dimensions of 610 × 340 pixels, and a spatial resolution of 1.3 m. To ensure the same number of spectral bands as PC, the last band of UP was removed. Both the scenes share the same seven classes of land cover. The names of the land cover classes and the number of samples for each class can be found in Table II. Furthermore, the pseudocolor images and ground-truth maps can be observed in Fig. 4.

3) S–H Dataset: The S–H dataset was acquired using the EO-1 Hyperion hyperspectral sensor, which captures 198 bands. The Shanghai dataset has dimensions of 1600 × 230 pixels and contains roads, buildings, plants, and the waters of the Yangtze River and Huangpu River. The Hangzhou dataset has dimensions of 590 × 230 pixels and includes roads, buildings, plants, West Lake, and the Qiantang River basin. Table III displays three common land cover classes utilized for the cross-scene classification task, while their pseudo-color images and ground-truth maps are visualized in Fig. 5.

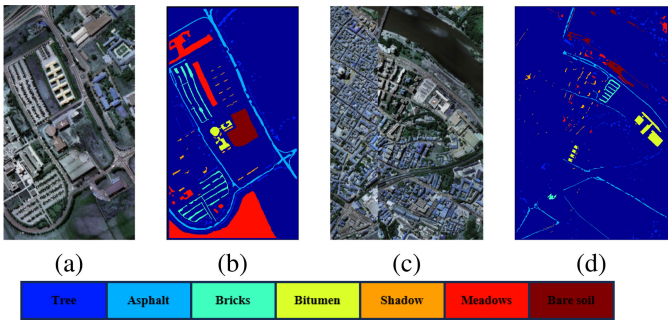


Fig. 4. Pseudocolor image and ground-truth map [17], [49] of the Pavia dataset. (a) Pseudocolor image of UP. (b) Ground-truth map of UP. (c) Pseudocolor image of PC. (d) Ground-truth map of PC.

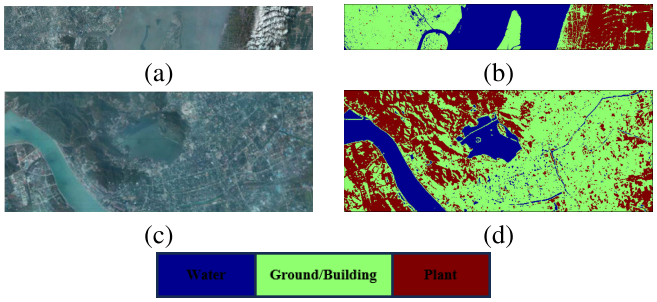


Fig. 5. Pseudocolor image and ground-truth map of S–H [17]. (a) Pseudocolor image of Shanghai. (b) Ground-truth map of Shanghai. (c) Pseudocolor image of Hangzhou. (d) Ground-truth map of Hangzhou.

TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE S–H DATASET

Class		Number of Samples	
No.	Name	Hangzhou (Source)	Shanghai (Target)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
Total		135700	368000

B. Experimental Setting

All the experiments in this study are conducted using the PyTorch platform. The flip augmentation and radiation augmentation were used in three datasets [17]. The optimization scheme used is mini-batch stochastic gradient descent (SGD) with a momentum of 0.9. The learning rate is adjusted using the formula: $\eta_\sigma = (\eta_0/(1+\alpha\sigma)^\beta)$, where σ represents the linear change in training progress from 0 to 1, η_0 is the base learning rate, α is set to 10, and β is set to 0.75, defaultly. In addition, the ℓ_2-norm regularization with a value of $5e^{-4}$ is used for weight decay to stabilize network training and mitigate overfitting. Finally, the batch size and the number of training epochs are set to 100. Each training epoch will contain 4000 iterations.

The structure of the feature extractor used in SERC is shown in Table IV. The Conv3d, Fc, and B here stand for 3-D convolution layer, full connection layer, and batch size, respectively. K is the number of classes. The flatten here is a dimensionality reduction function used to flatten a multidimensional array into a 1-D array. During initialization, Conv3d uses the Kaiming distribution and constant 1 initialization parameters, respectively, Fc in the multilayer perceptron uses

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TABLE IV
NETWORK STRUCTURE OF SERC

SERC	CNN
Input Dimension	B×7×7×d
Feature Extractor	3×3×3 Conv3d 3×1×1 Pool B×5×5×20
	3×3×3 Conv3d 3×1×1 Pool B×3×3×20
	2×1×1 Conv3d Dropout(0.1) Flatten B×962
Classifier head	Fc
Output Dimension	B×K

TABLE V
EXPERIMENTAL DETAILS OF COMPARISON ALGORITHMS

CLDA	Original	Select 180 samples from each category as the training set.
	Change	Treat all source domain samples as training set and the target domain samples as the test set and add results on Houston, Pavia and S-H.
TSTnet	Original	Select 5% samples of source domain as the training set.
	Change	Treat all source domain samples as the training set and the target domain samples as the test set and add results on Houston, Pavia and S-H.
SSWADA	Original	No results were provided on the Houston and Pavia.
	Change	Add results for Houston and Pavia.
CCGDA	Original	No test results were provided on the Pavia.
	Change	Add experimental results on the Pavia.

the normal distribution $N(0, 0.001)$, and the rest part uses the default parameter initialization in the library.

To reduce the size of the training set or to avoid class imbalance problems, some of the existing works may first filter the raw HSI and then use the filtered data as the training dataset, such as CLDA and TSTnet. However, different filtering strategies may lead to different model performance, which will greatly weaken the reliability of test results. Therefore, we treat all the SD samples as training sets and all the TD samples as test sets. To ensure fairness, all the comparison algorithms will follow this setting.

Since the classic algorithms (DAN, DANN, DSAN, MCD) do not provide official test results for cross-scene HSI classification, and the test results shown in existing literatures [17], [21], [23], [39] are often inconsistent. Therefore, we adopt a widely accepted approach: using ResNet18 as the backbone and a single layer fully connected layer as the classifier. The testing codes for them are sourced from here. For the other latest works, there are some differences in the experimental settings between the original works with our work. For example, CLDA uses only 180 SD samples from each category as the training set, while TSTnet uses only 5% of the SD samples. To ensure consistency between them with our proposed method, we will re-evaluate these algorithms. The experimental details of these comparison algorithms are shown in Table V. Specifically, if the original comparison algorithms did not perform any dataset filtering and have provided experimental results, we will use the original experimental results for comparison. If not, we will remove their data filtering operations and then test their performance across the three datasets for comparison.

TABLE VI
OVERALL CLASSIFICATION ACCURACY (%) WITH DIFFERENT NUMBERS OF NEIGHBORS λ_2 FOR THE PROPOSED SERC USING THE THREE EXPERIMENTAL DATASETS

Dataset	λ_2				
	0.1	0.3	0.5	0.7	0.9
Houston	78.3	80.5	**82.6**	81.3	81.3
Pavia	**89.6**	89.4	89.4	88.6	89.4
S-H	91.5	92.6	**93.7**	92.4	91.3
Mean	86.5	87.2	**88.5**	87.1	87.4

C. Parameter Tuning

In the proposed SERC, the adjustable hyperparameters include the regularization parameters: λ_0, λ_1, λ_2 and the base learning rate η_0. In this section, we will examine the impact of these parameters on the model's performance by systematically varying their values. The regularization parameters λ_0, λ_1, and λ_2 can take values from these sets $\{4, 8, 12, 16\}$, $\{0.2, 0.4, 0.6, 0.8\}$, and $\{0.1, 0.3, 0.5, 0.7, 0.9\}$. The base learning rate η_0 can be selected from the set $\{2e^{-5}, 2e^{-4}, 2e^{-3}, 2e^{-2}, 2e^{-1}\}$.

The parameter λ_0 represents the number of neighbors that determines the extent of captured global-spatial information. The parameter λ_1 is the tradeoff between the influence of the global-spatial and spectral information. Therefore, λ_0 and λ_1 play the role of regulating the contribution of global-spatial information to SERC simultaneously. Fig. 6 demonstrates the performance variation (represented by OA) of SERC across three experimental datasets when different combinations of λ_0 and λ_1 are used. The results demonstrate that SERC is less influenced by changes in λ_1, but will be largely influenced by λ_0. In summary, we set λ_0 to 12 for the Pavia and Houston datasets, and to 4 for the S–H dataset. In addition, we set λ_1 to 0.4 across all the datasets.

The hyperparameter λ_2 is the tradeoff between $\mathcal{L}_{\text{cdm}}$ and $\mathcal{L}_{\text{na}}$. This parameter controls the influence of the RC's guidance during training. Table VI presents the model's performance under different λ_2 values. The results indicate that the Houston dataset is highly sensitive to λ_2, followed by S–H, while Pavia exhibits relatively stable behavior. In this study, we select the parameter that yielded the highest average across the three datasets, ensuring that λ_2 is set to 0.5 for all the datasets.

The learning rate, denoted as η_0, is a key factor in the training process as it determines the magnitude of the gradient descent step. Consequently, it has a significant impact on the model's learning performance. In this study, we investigate the influence of various base learning rates on all the datasets. The performance results, measured by OA, are presented in Table VII. The highest is reached at a learning rate of $2e^{-2}$. The mean accuracy across three datasets at learning rates of $2e^{-5}$, $2e^{-4}$, $2e^{-3}$, $2e^{-2}$, and $2e^{-1}$ is 24.17%, 40.36%, 70.75%, 88.54%, and 0%, respectively. The highest accuracy is achieved at a learning rate of $2e^{-2}$.

In conclusion, for Houston and Pavia, the suboptimal parameters λ_0, λ_1, and λ_2 are set to 12, 0.4, and 0.5, respectively. For the S–H dataset, the values are 4, 0.4, and 0.5. The optimal base learning rate for all the datasets is $2e^{-2}$.

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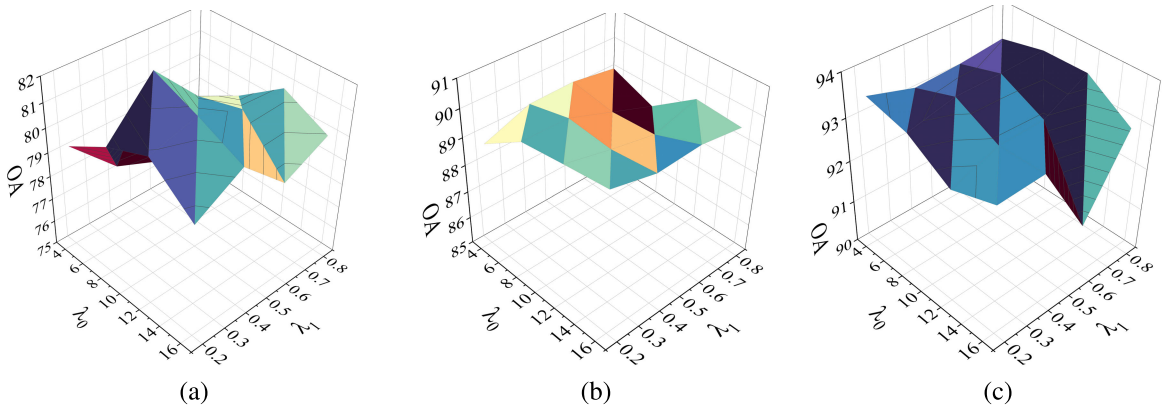


Fig. 6. Overall classification accuracy (%) under different combinations of λ_0 and λ_1 for the proposed SERC using the three experimental datasets. The X-axis and Y-axis are λ_0 and λ_1, respectively, and the Z-axis is the OA in TD. The η_0 used for all the datasets is 0.02. (a) Houston. (b) Pavia. (c) S–H.

TABLE VII

OVERALL CLASSIFICATION ACCURACY (%) WITH DIFFERENT BASE LEARNING RATES η_0 FOR THE PROPOSED SERC USING THE THREE EXPERIMENTAL DATASETS

Dataset	Base learning rate η_0				
	2e-5	2e-4	2e-3	2e-2	2e-1
Houston	10.1	18.3	58.5	**82.6**	0
Pavia	12.4	31.5	68.8	**89.4**	0
S-H	50.1	73.3	86.4	**93.7**	0
Mean	24.2	40.4	70.8	**88.5**	0

TABLE VIII

OPTIMAL PARAMETERS OF DIFFERENT DATASETS

	Houston	Pavia	S-H
η_0	0.2	0.2	0.2
λ_0	12	12	4
λ_1	0.4	0.4	0.4
λ_2	0.5	0.5	0.5
OA	82.6	89.4	93.7

For easy reading, we concluded the final selected value for the four parameters and show them in Table VIII. According to Table VIII, although we will select the optimal parameters for different datasets, which sounds like a parameter-strict approach, the best parameters across the three datasets are quite similar, with only a slight variation in the parameter λ_0. In fact, using the same parameters ($\lambda_0 = 12$) on the S–H dataset same as the Houston and Pavia datasets can still get a good performance (93.4%). This indicates that the same set of parameters ($\eta_0 = 0.2$, $\lambda_0 = 12$, $\lambda_1 = 0.4$, and $\lambda_2 = 0.5$) can achieve optimal or near-optimal results across all the three datasets, demonstrating that models using these parameters are robust across multiple datasets.

If there are no additional annotations, we adopt the parameters in Table VIII default in all the experiments. Note that in experiments of Fig. 6 and Tables VI and VII, except for the tested variable, the remaining parameters also adhere to this rule. For instance, in Table VI, λ_2 is varied, while λ_0, λ_1, and η_0 are kept at their default settings.

D. Ablation Analysis

The proposed SERC uses an RC to provide guidance to the CC, and the output of the RC is then used to further optimize the CC. Therefore, the RC and CC mutually reinforce each other. To demonstrate the effectiveness of this mechanism, performance curves during training were plotted for three types of classifiers: the CC without guidance (CC w/o), the CC with guidance (CC w/), and the RC.

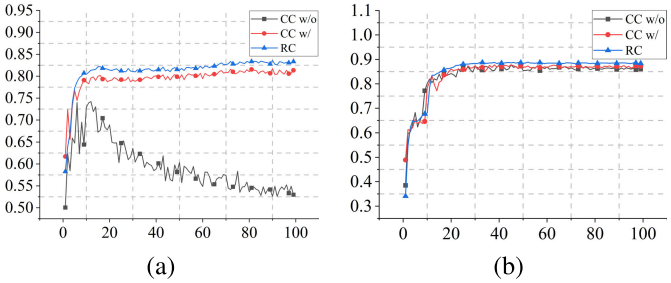


Fig. 7. Comparison between the CC without guidance (CC w/o), the CC with guidance (CC w/), and the RC. The horizontal axis is the iteration of training, and the vertical axis is the OA in TD. (a) Houston. (b) Pavia.

We conduct comparative experiments on the Houston and Pavia datasets, and the results are depicted in Fig. 7. The impact of the guidance provided by the RC varies across different datasets. As shown in Fig. 7(a), the CC (w/o) initially achieves high accuracy, but quickly experiences a significant decrease in the Houston dataset (gray line), which is mainly due to the rapid overfitting on the SD. However, this phenomenon was not observed in the training curves of the CC (w/), proving the guidance from RC can effectively mitigate the overfitting issue (red line). Furthermore, the RC (blue line) outperforms the CC (w/) due to its incorporation of global-spatial information, which enhances its performance.

It indicates that although RC is constructed based on the knowledge of the CC (w/), its predictions are the aggregation of their coarse predictions. But the fusion of global-spatial information can indeed make the aggregated refined predictions surpass the coarse predictions. Furthermore, the refined predictions can be transferred to pseudolabels, which can inversely feed back to the CC, which helps prevent its overfitting on the SD. However, we also note that this guidance cannot yield equally significant effects on all the datasets. As shown in Fig. 7(b), on the Pavia dataset, although the RC still demonstrates superior performance compared with both the types of CC, the promotion is not as pronounced as observed in the Houston dataset. The main reason for the

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TABLE IX
OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THREE PUBLIC DATASETS

	Houston	Pavia	S-H	Mean
SERC (w/o CDM)	80.3	88.7	92.7	87.2
SERC (w/ CDM)	**82.6**	**89.4**	**93.7**	**88.5**

difference in effectiveness lies in whether the dataset is susceptible to overfitting on the SD, which refers to the severity of negative transfer. In the Houston dataset, this negative transfer phenomenon (overtraining leads to performance degradation) is extremely serious. But this phenomenon is not obvious in the Pavia dataset. This difference proves that the proposed SERC is more effective in the cross-scene classification task with large CDS.

To evaluate the impact of CDM on the model, we assess the model's performance with and without incorporating CDM, which is represented by SERC (w/ CDM) and SERC (w/o CDM). Table IX demonstrates the substantial performance improvement achieved by incorporating CDM into the model. This improvement is particularly pronounced in the Houston dataset, where SERC without CDM achieves 80.32%, while SERC with CDM achieves 82.56%, resulting in a 2.24% enhancement. In addition, CDM leads to improvements of 0.66% and 1.01% on the Pavia and S–H datasets, respectively. On average, the model achieves an improvement of 1.31%.

E. Performance on Cross-Scene HSI Classification

Table X presents the experimental results obtained on the Houston dataset. The Houston2013 dataset serves as the SD and the Houston2018 dataset serves as the TD. Our proposed SERC outperforms all other methods, achieving an OA of 82.6% and a KC of 70.5. SERC achieves the state-of-the-art (SOTA) performance among these methods, surpassing CCGDA, which is the best performing of the existing method, by 5.4%. Fig. 8 presents the classification maps of all the methods on the Houston dataset, illustrating the prediction results. Notably, the prediction result of SERC exhibits a higher level of consistency, devoid of notable noise, and with clear category borders. For instance, in the region highlighted by the purple box in Fig. 8(j), the existing methods frequently misclassify nonresidential buildings as residential buildings. In our proposed SERC, while there are still instances of misidentification, the occurrence of such errors is significantly reduced.

Table XI displays the experimental results obtained on the Pavia dataset. The SD is UP, and the TD is PC. Our proposed SERC outperforms all other methods, achieving an OA of 89.4% and a KC of 87.2. SERC achieves the SOTA performance among these methods, surpassing CLDA, the best-performing existing method, by 11.9%. Notably, the CLDA, TSTnet, and SSWADA algorithms exhibit poor performance on Class 4 (Bitumen) in the Pavia dataset, with scores of 0%, 47.7%, and 26.6%, respectively, which negatively impact their overall OA performance. This is because the Bitumen class has a proportion of 3.4% in the SD and 18.5% in the TD, creating a significant discrepancy that prevents models trained on the SD from adapting to the TD scenario.

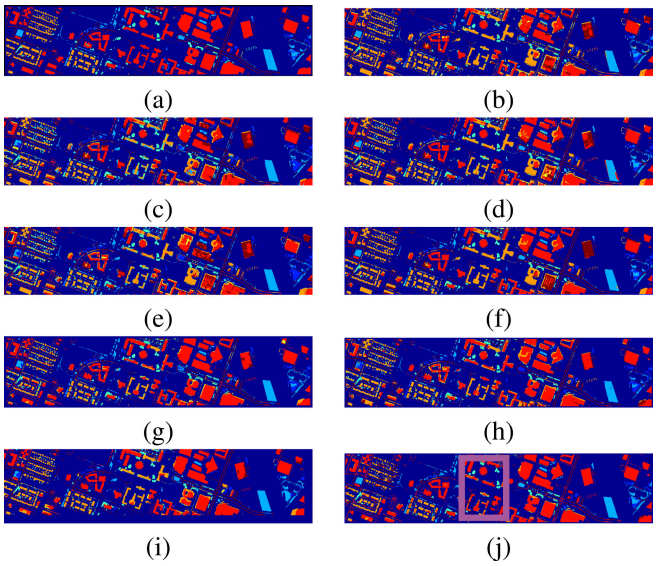


Fig. 8. Data visualization of classification maps for target scene Houston 2018 data obtained with different methods including (a) GT, (b) DAN, (c) DANN, (d) DSAN, (e) MCD, (f) CLDA, (g) TSTnet, (h) SSWADA, (i) CCGDA, and (j) SERC.

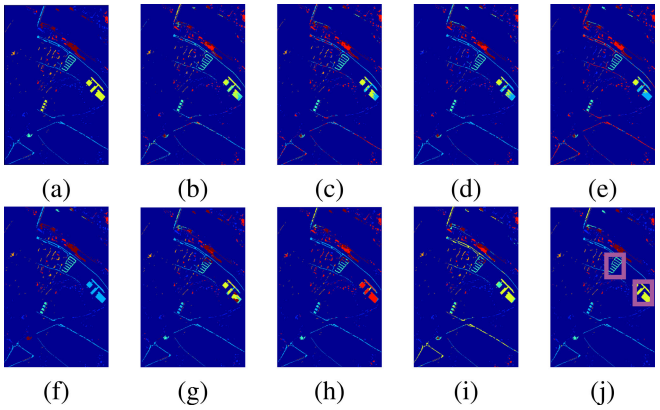


Fig. 9. Data visualization of classification maps for target scene PC data obtained with different methods including (a) GT, (b) DAN, (c) DANN, (d) DSAN, (e) MCD, (f) CLDA, (g) TSTnet, (h) SSWADA, (i) CCGDA, and (j) SERC.

In contrast, the proposed SERC algorithm avoids this issue, highlighting the significance of our method. Fig. 9 shows the classification maps of all the methods for the Pavia dataset. The results achieved by SERC outperform other methods. Specifically, in the purple box region of Fig. 9(j), the existing methods often misclassify bricks and bitumen. In contrast, SERC effectively avoids this problem, resulting in improved detection of bricks and bitumen with smoother and noise-free boundaries.

Table XII shows the experimental results in the S–H dataset. The SD is Hangzhou, and the TD is Shanghai. Our SERC gets the second-best performance on the S–H dataset, which is 93.7%, only behind SSWADA (94.5%). Fig. 10 shows the classification maps of existing methods and our proposed SERC in the S–H dataset. The primary reason for this is the presence of numerous scattered points in the Shanghai scene of the S–H dataset, indicating rapid scene changes. While the SERC model leverages neighbor information to improve prediction accuracy in most cases, this reliance also hampers

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TABLE X

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018 DATA, WHERE THE VALUE IN BOLD IS THE BEST

Class	Classification algorithms								
	DAN	DANN	DSAN	MCD	CLDA	TSTnet	SSWADA	CCGDA	SERC
1	80.1	41.2	76.0	46.2	88.6	85.6	93.9	85.9	77.8
2	80.2	72.8	86.2	75.2	64.9	85.1	67.6	64.3	85.1
3	57.0	47.5	52.8	36.2	69.7	71.9	66.5	70.2	78.3
4	1.0	2.2	1.2	0.6	95.5	100	100	81.9	87.2
5	39.5	32.3	31.8	30.7	93.0	73.8	85.0	88.9	70.8
6	91.1	95.3	87.3	92.9	60.9	84.6	72.2	82.4	90.0
7	37.2	47.6	50.0	43.0	85.9	51.6	80.0	63.3	53.9
OA (%)	63.6	61.5	63.3	57.5	68.6	79.0	74.3	78.1	**82.6**
KC (κ)	51.7	49.6	50.8	45.6	55.9	66.2	61.7	66.1	**70.5**

TABLE XI

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE PC DATA, WHERE THE VALUE IN BOLD IS THE BEST

Class	Classification algorithms								
	DAN	DANN	DSAN	MCD	CLDA	TSTnet	SSWADA	CCGDA	SERC
1	62.8	84.0	64.6	93.6	92.7	80.6	69.6	87.3	90.4
2	94.4	64.3	69.6	21.0	99.8	91.4	86.9	62.0	92.1
3	51.0	54.3	51.6	53.9	79.1	66.6	86.0	98.5	81.8
4	91.4	100	93.2	79.9	0	47.7	26.6	73.3	88.5
5	99.5	98.9	99.2	100	99.9	73.7	84.2	99.6	91.6
6	33.2	27.2	38.9	19.2	99.6	82.3	84.9	76.2	83.2
7	82.7	80.6	88.5	85.1	93.7	80.1	64.7	74.4	91.5
OA (%)	70.7	62.6	65.7	49.8	77.5	75.6	68.3	77.4	**89.4**
KC (κ)	66.6	58.4	61.2	45.9	72.7	68.3	62.4	73.2	**87.2**

TABLE XII

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE SHANGHAI DATA, WHERE THE VALUE IN BOLD IS THE BEST

Class	Classification algorithms								
	DAN	DANN	DSAN	MCD	CLDA	TSTnet	SSWADA	CCGDA	SERC
1	100	100	100	100	93.6	91.0	95.9	93.9	93.3
2	63.9	77.5	90.9	92.3	88.2	74.9	91.7	90.3	91.2
3	80.8	86.3	84.8	82.6	99.1	99.1	97.9	100	99.1
OA (%)	72.9	84.7	92.1	92.0	92.4	85.9	**94.5**	93.0	93.7
KC (κ)	63.3	78.1	88.2	88.1	88.4	78.8	**91.5**	89.4	90.3

TABLE XIII

OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THREE PUBLIC DATA, WHERE THE VALUE IN BOLD IS THE BEST

	Houston	Pavia	S-H	Mean
CLDA	68.6	77.5	92.4	79.5
TSTnet	79.0	75.6	85.9	80.2
SSWADA	74.3	68.3	94.5	79.0
CCGDA	78.1	77.4	93.0	82.8
SERC	**82.6**	**89.4**	**93.7**	**88.6**

its ability to handle rapidly changing scene information. This limitation requires further exploration in future work.

Table XIII gives the average result in three datasets. We select four best existing methods for comparison. Our SERC achieves very excellent performance. The average performance is 5.8% higher than the best existing methods. The aforementioned results demonstrate the exceptional performance of the proposed SERC method across the three datasets.

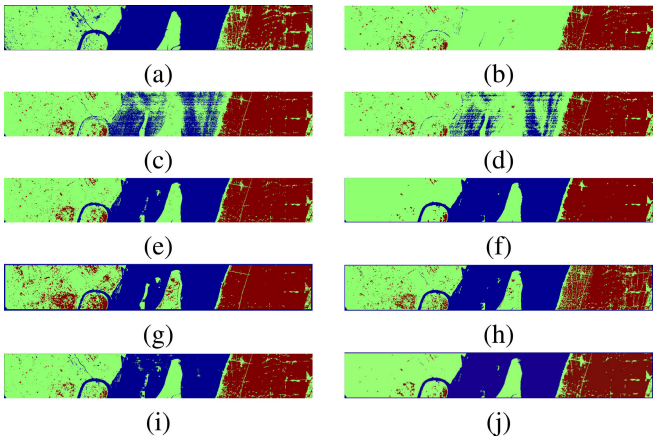


Fig. 10. Data visualization of classification maps for target scene Shanghai data obtained with different methods including (a) GT, (b) DAN, (c) DANN, (d) DSAN, (e) MCD, (f) CLDA, (g) TSTnet, (h) SSWADA, (i) CCGDA, and (j) SERC.

F. Computational Complexity

In this section, we will analyze the computational complexity of the proposed SERC. All the comparison algorithms use the default settings from the original papers. We will compare their floating-point operations (FLOPs) and number of deep models' parameters. For MCD and CLDA, since their models consist of several subnetworks, the parameters recorded here

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TABLE XIV
COMPARISON OF COMPUTATIONAL COMPLEXITY

	DAN	DANN	DSAN	MCD	CLDA	TSTnet	SSWADA	CCGDA	SERC
FLOPs	19.53 M	19.69 M	19.53 M	19.69 M	3.37 M	8.50M	12.35 M	80.09 M	3.70 M
params	11.48 M	11.63 M	11.48 M	11.63 M	0.18 M	5.33M	7.05 M	4.25 M	0.02 M

is the total for all the subnetworks combined. Due to slight variations in model structures across different datasets, we will standardize the test dataset to Houston.

As shown in Table XIV, since DAN, DANN, DSAN, and MCD all use ResNet18 as their backbone, their FLOPs and number of parameters are relatively similar. Among the other comparison algorithms, CCGDA has the highest FLOPs due to its inclusion of three subnetworks: the generator, classifier, and discriminator. In terms of FLOPs, CLDA performs the best, with the lowest value of only 3.37M. The FLOPs of our proposed model, SERC, are very close to those of CLDA, reaching a suboptimal result of 3.70M. In terms of the number of parameters, SERC has only 0.02M, which is significantly lower than all the other algorithms.

Furthermore, the computational complexity of finding the nearest N neighbors must be considered. This complexity includes the cost of computing distances and the sorting algorithm. Suppose there are n_t samples in the TD and the dimension of the obtained feature is d. The complexity of calculating the cosine similarity and Euclidean distance is $O(dn_t^2)$ and $O(dn_t^2)$, and the overall computational complexity is $O((d+2)n_t^2)$. Then we call the torch.sort() function in PyTorch to filter the nearest neighbors for each sample, whose computational complexity is $O(n^2 \log(n))$. Therefore, the overall complexity is $O((d+2)n_t^2 + n^2 \log(n))$.

The above experiments demonstrate that our proposed model is both simple and lightweight. However, we recognize that the nearest neighbor search can be burdensome with larger datasets, which will need improvement in the future work.

G. Stability Evaluation

This section aims to demonstrate that the RC proposed in this article can enhance the CC, no matter the performance of the CC is high or low. To simulate the situation when the CC performs badly, we reduce the depth of the CC model to reduce its feature learning ability. Specifically, we tried three backbones with different structure and test the performance of the model under these backbones. The test dataset is the Houston dataset, and the SD and TD were Houston2013 and Houston2018, respectively. Similar to Section IV-D, the model chosen for comparison is the CC without guidance (CC w/o), the CC with guidance (CC w/), and the RC.

As shown in Table XV, the performance of CC (CC w/o) will directly affect the performance of RC. The higher the performance of CC (CC w/o), the higher the performance of RC. The lower, the vice versa. However, in all the above cases, both the CC with guidance (CC w/) and the RC trained with refined pseudolabels can achieve better results than the CC (CC w/o). This proves that the refined pseudolabels $\hat{p}_i$ generated in this article can play a positive role, no matter the performance of the CC (CC w/o) is good or poor. The above experiment shows that the RC in this article has high stability.

TABLE XV
COMPARISON OF PERFORMANCE UNDER DIFFERENT CC

	1	2	3
Structure of backbone	3×3×3 Conv3d 3×1×1 Pool Dropout(0.1) Flatten()	3×3×3 Conv3d 3×1×1 Pool 3×3×3 Conv3d 3×1×1 Pool Dropout(0.1) Flatten()	3×3×3 Conv3d 3×1×1 Pool 3×3×3 Conv3d 3×1×1 Pool 2×1×1 Conv3d Dropout(0.1) Flatten()
Coarse classifier without guidance (CC w/o)	74.9	74.5	75.7
Coarse classifier with guidance (CC w/)	75.0	77.9	79.8
Refined classifier	77.3	81.2	82.6

H. Feature Visualization

1) Visualization of Source and Target Features: The major challenge of the cross-scene HSI classification task is distributional inconsistency between the SD and TD. The purpose of this section is to demonstrate that the features extracted by the SERC model from the SD and TD are quite similar. To be specific, we input samples from the same category of two domains into the trained network. Then, the t-distributed stochastic neighbor embedding (t-SNE) dimensionality reduction technique was used to visualize their features. If features of samples from the same category in two domains tend to overlap, then the model can be proved to efficiently mitigate the issue of distribution inconsistency. Taking the second category (Asphalt) from the Pavia dataset as an example, this category has many samples, and there is significant intraclass variation. The red and blue dots represent samples from the SD and TD, respectively.

As depicted in Fig. 11, the features extracted from the SD and TD by the existing methods are not fully fused. For instance, models such as DAN, DANN, DSAN, MCD, and SSWADA exhibit significant domain gaps between the SD and TD features, which may lead to misclassification of samples from the TD. In contrast, models such as CLDA, TSTnet, and CCGDA show considerable improvement, with some overlap between the source and target features; however, a clear boundary still exists. In our model, the features from the SD and TD completely overlap, demonstrating the best performance and validating the effectiveness of the proposed method.

2) Visualization of Class Distribution Shift: This section aims to prove that the proposed SERC can effectively mitigate

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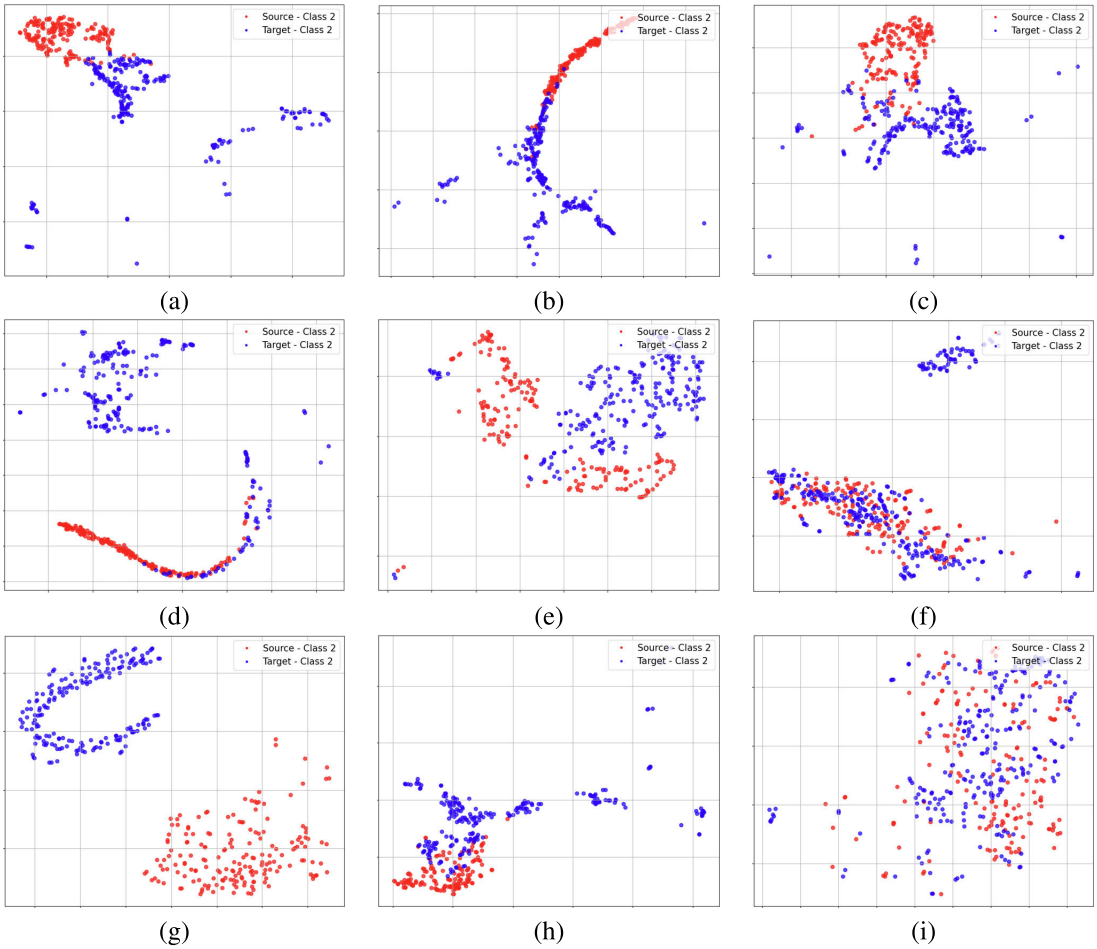


Fig. 11. t-SNE visualization of SD and TD features obtained with different methods including (a) DAN, (b) DANN, (c) DSAN, (d) MCD, (e) CLDA, (f) TSTnet, (g) SSWADA, (h) CCGDA, and (i) SERC.

misclassification problems caused by CDS compared with existing methods. To be specific, we compare the classification performance of SERC and existing algorithms on categories with significant CDS. Taking the Pavia dataset as an example, the fourth category (bitumen) has a proportion of 3.4% in the SD and 18.5% in the TD. This will cause the classification model trained on the SD to easily overlook the fourth category, making it difficult for them to adapt to the TD scenario. To illustrate it, we visualize the features of the TD samples and color them using two types of labels: one using the ground-truth labels and the other using the model's predicted labels. Different categories are distinguished using different colors. By comparing them, we can easily analyze which categories the model misclassified.

SSWADA, a representative UDA approach in cross-scene HSI classification problem, is used as the comparison algorithm in this study. Fig. 12(a) and (b) shows the visualization of features obtained using the SSWADA model, with feature points colored according to the ground-truth labels and predicted values, respectively. In contrast, Fig. 12(c) and (d) presents the features obtained using the SERC model, with the results colored according to the ground-truth labels and the predicted values, respectively. It can be seen that although the performance is strong in most categories, the SSWADA model experiences numerous misclassifications in the fourth category (colored in red) of the Pavia dataset.

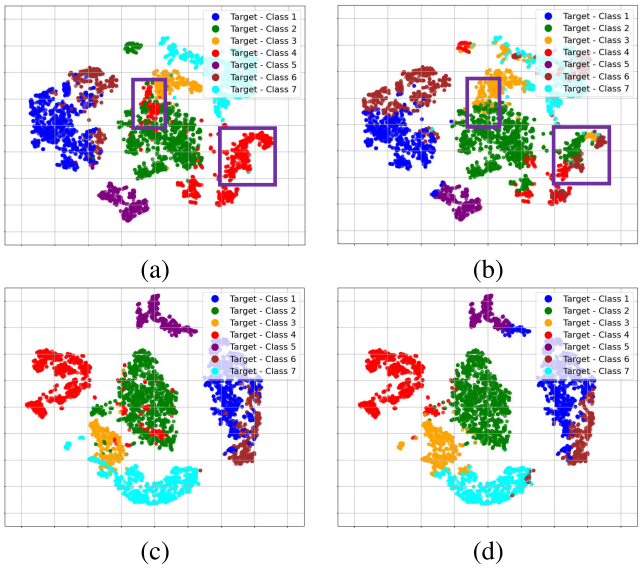


Fig. 12. t-SNE visualization of TD features. (a) Features output by SSWADA, colored using ground-truth labels. (b) Features output by SSWADA, colored using predicted labels. (c) Features output by SERC, colored using ground-truth labels. (d) Features output by SERC, colored using predicted labels.

To enhance readability, we have highlighted these feature points with purple boxes. In contrast, our proposed SERC model does not exhibit this issue and demonstrates high classi-

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fication accuracy across all the categories. These experiments indicate that the proposed method is more robust to CDS compared with the existing methods.

V. Conclusion

This article focuses on the issue of UDA in the cross-scene HSI classification task. The main contribution of this article is the proposed SERC, which is especially effective when large CDS exists. The proposed SERC constructs a memory module to fuse global-spatial and spectral information simultaneously and uses the neighborhood aggregation technique to generate the refined predictions from coarse predictions. Simultaneously, incorporating CDM into the model can help it focus on specific categories, resulting in a learned class distribution that closely resembles the TD. This approach effectively mitigates the problem of CDS. Experiments show that our SERC has achieved excellent results on the three public datasets: Houston, Pavia, and S–H.

However, a drawback of the proposed SERC is the substantial memory overhead used to store the memory module when handling large-sized datasets. In the follow-up research, further optimization can be carried out to reduce the required storage space.

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